

Exploratory Data Analysis Report

Dataset: Titanic Dataset

1. Missing Values Analysis

- Age column contains significant missing values (~177).
- Cabin column has majority missing (~687), not reliable for modeling.
- Embarked has very few missing values.
- Age imputation required before model building.

2. Outlier Detection (Fare)

- Fare distribution is right-skewed.
- Extreme high fare values present (300–500+).
- Log transformation may help improve modeling performance.

3. Correlation Analysis

- Survived & Fare → Positive correlation (~0.26).
- Survived & Pclass → Negative correlation (~-0.34).
- Strong negative correlation between Pclass & Fare (~-0.55).
- SibSp & Parch moderately correlated (~0.41).

4. Business Insights

- Higher class passengers had better survival chances.
- Wealth (Fare paid) influenced survival probability.
- Lower class passengers faced disadvantage.
- Cabin feature may be dropped due to excessive missing data.